

DATA ANALYTICS LEVEL III

Institutional Competency Assessment

Part II — Comprehensive Performance Task (Cores 1, 2, 4)

Candidate Copy

About this Assessment

This document is one part of the Institutional Competency Assessment for Data Analytics Level III. The full institutional assessment determines whether you are COMPETENT across the three Units of Competency listed below.

Unit of Competency (Core)	Learning Outcomes Assessed	Primary Tool
Core 1 — Apply Basic Statistical Concepts and Analysis	LO1 Apply key basic statistical concepts; LO2 Apply basic statistical analysis	Microsoft Excel
Core 2 — Prepare Data Sets	LO1 Import data; LO2 Clean data; LO3 Filter data	SQL (VS Code + SQLTools + SQLite), Excel Power Query
Core 4 — Prepare Data Visualization	LO1 Apply data visualization concepts; LO2 Create data visualization; LO3 Publish and share report	Microsoft Power BI Desktop + Service

The full institutional assessment has **TWO** parts:

Part	What it is	Time	Weight
Part I — Written Knowledge Test	50 items covering Cores 1, 2, and 4	1 hour	40%
Part II — Comprehensive Performance Task	End-to-end project (Excel, SQL, Power BI)	Administered across the capstone window (Days 13–15)	60%

This document covers **Part II only — the Comprehensive Performance Task**.

Passing standard: 80% overall AND all CRITICAL criteria met = COMPETENT | Otherwise = NOT YET COMPETENT

Materials and Tools You Will Use

- Microsoft Excel (Microsoft 365 or Excel 2021) with the Data Analysis ToolPak enabled

- VS Code + SQLTools extension + SQLTools SQLite driver
- Microsoft Power BI Desktop (free) and a Power BI Service account (free) for publishing
- Assessment data files (provided on the LMS): InstitutionalAssessment_transactions.csv, InstitutionalAssessment_products.csv, InstitutionalAssessment_stores.csv

Specific Instructions to the Candidate

1. This task is OPEN-RESOURCE (you may use your own notes and handouts) but must be entirely YOUR OWN WORK. Plagiarised or shared outputs are rated Not Yet Competent.
2. Save each deliverable using the exact filenames given and replace [YourName] with your full name.
3. Your FINAL submission is your GitHub repository (see “Final Submission via GitHub” below). Push all deliverables and submit your repo URL to the LMS before the posted deadline. Late, missing, or inaccessible repositories cannot be rated.
4. Observe OSH and data-privacy practices: take screen breaks, and do not share or expose any personal data in your outputs or in your public repository.
5. If anything is unclear, raise it with your assessor before proceeding.

Comprehensive Performance Task

Weight: 60% | Administered across the capstone window (Days 13–15)

Business Scenario

Bahaghari Retail Co. operates several convenience stores across Luzon, Visayas, and Mindanao. Management has given you one year of raw transaction data spread across three files and wants an analysis that shows how the business is performing and where to focus next. You must take the raw data all the way to a published, interactive dashboard — demonstrating the three core competencies assessed (Cores 1, 2, and 4).

The Data (three raw files, provided on the LMS)

File	Key Columns	Notes (the data is deliberately 'messy')
InstitutionalAssessment_transactions.csv	transaction_id, date, store_id, product_id, units, unit_price, total_amount	Contains duplicate rows, some blank units, and dates in mixed formats.
InstitutionalAssessment_products.csv	product_id, product_name, category, unit_cost	Category text is inconsistent (e.g., 'beverages', 'Beverages ', 'BEVERAGES').
InstitutionalAssessment_stores.csv	store_id, store_name, region	A few store_name values have leading/trailing spaces; one region is NULL.

What You Must Produce (mapped to Cores 1, 2, and 4)

Part A — Core 1: Descriptive Statistics (Excel)

6. Compute mean, median, and mode for total_amount and for units.
7. Compute range, variance, standard deviation, and the quartiles (Q1, Q2, Q3) for total_amount.
8. Create a histogram of total_amount.
9. Write 3–4 sentences interpreting the centre and spread (e.g., what the mean vs. median tells you).

Part B — Core 2: Prepare the Data (SQL and/or Power Query)

10. Import all three files into SQLite (via VS Code + SQLTools) and/or Excel Power Query.
11. Clean the data: remove duplicate transactions; TRIM spaces; standardize category text to one consistent case; handle the blank units and the NULL region sensibly; correct any wrong data types.
12. Filter the transactions to the most recent full year (or as the assessor specifies).
13. Join the three tables into ONE analysis-ready dataset (transaction + product + store details).
14. Submit your SQL script (.sql) AND the cleaned Excel workbook.

Part C — Core 4: Visualize and Publish (Power BI)

15. Import the cleaned data into Power BI Desktop and build relationships (star schema: transactions = fact; products and stores = dimensions).
16. Create at least 2 DAX measures (e.g., Total Sales, Average Order Value).
17. Build at least 5 visuals: a KPI card, a bar/column chart (by region), a line chart (monthly trend), a composition chart (by category), and one more of your choice.
18. Add a slicer and one interactivity feature (drill-through or bookmark navigation).
19. Publish the report to the Power BI Service and submit the .pbix plus the published link or a screenshot.

Final Deliverable — Executive Summary

20. Write a 1-page executive summary: 3+ evidence-based insights from your Power BI analysis and one recommendation for Bahaghari Retail's management.
21. Complete the Data Quality (QA) checklist before submitting.

Files to Submit (placed in your GitHub repository)

All files below go into your GitHub repository using the folder structure shown in the “Final Submission via GitHub” section.

Deliverable	Filename
Statistics workbook (Part A)	IA_Statistics_[YourName].xlsx
SQL script (Part B)	IA_Preparation_[YourName].sql
Cleaned + joined workbook (Part B)	IA_PreparedWorkbook_[YourName].xlsx
Power BI report (Part C)	IA_Dashboard_[YourName].pbix
Published link / screenshot (Part C)	IA_PublishedLink_[YourName].txt or .png

Executive summary	IA_ExecutiveSummary_[YourName].pdf or .docx
Project README	README.md (in repo root)

How Your Performance Task Will Be Assessed (100 points)

Your assessor will rate your deliverables against these components. Criteria marked CRITICAL must be satisfied to be rated Competent.

Component	Points	What is assessed
Part A — Core 1: Descriptive Statistics (Excel)	25	Correct mean/median/mode, range, variance, standard deviation, and quartiles for the required columns; histogram present; written interpretation of centre and spread.
Part B — Core 2: Prepare Data (SQL / Power Query)	30	All 3 source files imported; duplicates removed; text trimmed and case-standardized; NULLs handled; data types corrected; required filter applied; tables joined into one analysis-ready dataset. .sql script and cleaned workbook submitted.
Part C — Core 4: Visualization (Power BI)	35	Data imported and relationships built (star schema); at least 2 DAX measures; 5+ appropriate visuals incl. KPI card, bar, line, and composition chart; slicer + one interactivity feature (drill-through/bookmark); report published to Power BI Service.
Executive Summary & Data Quality	10	One-page executive summary with 3+ evidence-based insights and a recommendation; QA checklist completed; professional, consistent formatting across deliverables.

CRITICAL criteria (must all be met): correct descriptive statistics for the required columns (Part A), cleaned & joined dataset (Part B), a published Power BI report with relationships (Part C), AND a GitHub repository that follows the required structure and is accessible to the assessor.

Final Submission via GitHub

Your final submission for this performance task is a GitHub repository containing every deliverable. The published Power BI report (Power BI Service) plus the GitHub repo together form your complete submission.

1. Create the repository

- Sign in to GitHub (<https://github.com>). Create a NEW repository.
- **Repository name:** ****data-analytics-LIII-IA-[YourName]**** (lowercase, dashes — replace [YourName], e.g., data-analytics-LIII-IA-juan-dela-cruz).
- **Visibility:** ****Public**** is preferred. If you make it ****Private****, you **MUST** invite your assessor as a Collaborator (Settings → Collaborators) before the deadline, otherwise the submission is inaccessible and cannot be rated.
- Initialize with a README so the repo is not empty on creation.

2. Required folder structure

Use exactly these folders so the assessor can navigate quickly:

```
data-analytics-LIII-IA-[YourName]/
├── README.md
├── 01_statistics/
│   └── IA_Statistics_[YourName].xlsx
├── 02_data_preparation/
│   ├── IA_Preparation_[YourName].sql
│   └── IA_PreparedWorkbook_[YourName].xlsx
├── 03_visualization/
│   ├── IA_Dashboard_[YourName].pbix
│   └── IA_PublishedLink_[YourName].txt
└── 04_executive_summary/
    └── IA_ExecutiveSummary_[YourName].pdf
```

3. README.md (must include)

- Your full name and the date of submission
- One-paragraph project overview (the Bahaghari Retail scenario, in your own words)
- A short description of each folder's contents
- The link to your PUBLISHED Power BI report on the Power BI Service
- Tools used (Excel version, SQLite, Power BI Desktop) and any setup notes needed to open the .pbix
- A short statement that the work is your own (academic integrity)

4. Push your work

From your project folder on your computer:

```
git init
git add .
```

```
git commit -m "IA: initial commit – project structure and README"
git branch -M main
git remote add origin https://github.com/<your-username>/data-analytics-LIII-IA-[YourName].git
git push -u origin main
```

- Make several commits as you work (e.g., one per Part) — a single big “final dump” commit does not show authentic, incremental work and may be flagged.
- **Large files:** .pbix and .xlsx are fine on GitHub; if any file exceeds 100 MB, ask your assessor about Git LFS BEFORE pushing.

5. Submit your repository URL

- Once your final commit is pushed, copy your repo URL (e.g., [https://github.com/<your-username>/data-analytics-LIII-IA-\[YourName\]](https://github.com/<your-username>/data-analytics-LIII-IA-[YourName])).
- Submit the URL via the LMS submission slot AND email it to your assessor before the deadline.
- Do NOT delete, rename, or make the repository private after submission. It must remain accessible to the assessor until your competency decision is recorded.

Pre-submission checklist

- ☐ Repo created with the correct name and the required folder structure
 - ☐ All six deliverables placed in the correct folders, using the correct filenames
 - ☐ README.md complete and includes the published Power BI link
 - ☐ Power BI report published to the Power BI Service and the link works
 - ☐ Repo is Public, OR Private with the assessor invited as Collaborator (accepted)
 - ☐ Repo URL submitted via the LMS before the deadline
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Candidate Declaration

I certify that the work I submit for this performance task is entirely my own.

Candidate Name:

Signature / Date:
